Data Preparation and Validation

Project: Global Olympic Sports Performance Analysis

Preparation workflow

- Inspect all four source CSV files and record row counts, field names and data types.
- Standardize country/NOC identifiers and year formats before creating relationships or joins.
- Confirm whether each file is athlete-level, country-year-level, sport-level or investment-level.
- Check duplicates and missing values; distinguish true zeros from null values.
- Confirm numeric fields such as total_medals, sports_investment, athletes_sent, gold, silver, bronze and medal_efficiency_pct are numeric.
- Validate region categories against a documented mapping, particularly historical or mixed teams.
- Recalculate derived metrics independently and compare with supplied fields.

Relationship keys

Potential keys discussed for this project include year, country and noc_code. Use only keys that are present and unique at the correct grain in each source. Avoid joining multiple fact tables on incomplete keys, as this can multiply rows and inflate totals.

Efficiency definition

If the agreed definition is medal efficiency percentage = total_medals / athletes_sent × 100, confirm that both measures refer to the same country, time period and unit of analysis. Exclude or flag rows where athletes_sent is zero or missing. Do not assume that a field with a similar name uses this exact formula until verified.

KPI verification

Previously discussed KPI examples (such as 910 participation records, 60 sport-category entries, 18 nations, or 33.2% efficiency) are provisional and must be recalculated from the actual supplied datasets before they are presented as project results.

Validation checklist

- Compare totals before and after joins.
- Check unique country/NOC and year counts.
- Spot-check selected country-year records against source rows.
- Confirm all chart filters respond as intended.
- Document any excluded rows and the reason for exclusion.